



Exploring Flora as a Nature-Based Solution to Combat Beach Erosion in Algeciras



Beaches of Algeciras (Pérez, 2020)

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1. Introduction

Coastal communities worldwide face a growing threat: beach erosion. This complex issue, driven by both natural processes like wind and wave action and exacerbated by human activities, is further intensified by climate change. Rising sea levels and intensifying storms contribute to coastal flooding and erosion, jeopardizing ecosystems, infrastructure, and the very fabric of coastal communities.

The case of Algeciras, Spain, serves as a stark reminder of these consequences. Two beaches, El Chorruelo and Los Ladrillos, have already been lost, and El Rinconcillo faces a similar fate due to port expansion efforts. While traditional methods like beach nourishment and rehabilitation are used for coastal management, they often exacerbate existing problems, as the Salvemos Rinconcillo group reports.

This project seeks to explore and implement a flora restoration project as a nature-based solution for beach erosion in Algeciras: utilizing flora. Plant life plays a crucial role in protecting sandy coastlines, offering natural defense against storms and flooding. However, climate change poses a significant threat to coastal biodiversity, diminishing this vital protection. By investigating how flora can be utilized effectively, this project aims to:

1. Document and understand the existing plant life in the study area.
2. Identify current practices and policies used to address beach erosion.
3. Develop strategies to integrate nature-based solutions (using flora) into existing coastal management practices.

Through interviews with coastal management and environmental agencies, transect walks with experts, and collaborative discussions with stakeholders, this project will explore the potential of utilizing flora as a sustainable and effective solution to mitigate beach erosion in Algeciras, ensuring the preservation of both beaches and biodiversity. Including the final report, the final deliverable will include organized focus groups with interested stakeholders, environmental agencies, and university faculty members to incentivize a

broader discussion of the effectiveness of a nature-based solution to properly preserve both beaches and biodiversity in Algeciras.

2. Background

2.1. Climate Change and Beach Erosion

2.1.1 Global Context

Climate change casts a long shadow over communities worldwide, especially coastal communities. The increase in greenhouse gasses traps heat in the atmosphere, warming the planet and its surfaces. This translates to rising ocean temperatures, increasing sea levels, and acidification, human-driven increased levels of carbon dioxide in the atmosphere are dissolving into the ocean which makes the pH decrease and the ocean becomes more acidic, which constitutes a triple threat harming marine life and disrupting the coastal ecosystems. These extreme climate conditions and spike in storm activity have led to severe flooding, one of the main causes of this recede (The Climate Reality Project, 2019). Beaches play a pivotal role in preserving both ecosystems and coastal structures, serving as a buffer that protects against erosion and supports diverse habitats that are essential for the sustainability of coastal environments.

The United States Environmental Protection Agency reports a direct correlation between the sea levels and annual temperature rise in the Iberian Peninsula and increased storm frequencies and wave heights which will consequently increase coastal flooding and erosion (Environmental Protection Agency, 2021). The action of storms is a very important natural factor causing shoreline erosion because it generally triggers beach flattening, erosion escarpments on beaches and dunes, and overwash processes, especially on steeper beaches due to their higher susceptibility to changes in wave regime (Cooper et al., 2004).

Sea levels and increased storm frequencies are not the only natural factors that influence shoreline evolution. Natural factors, such as winds, waves, and coastal settings play a crucial role in shoreline changes. Easterly wind (Levante) heavily influences aeolian sediment transport, while westerly winds (Poniente) hold great significance in wave generation, especially during winter storm conditions (Benavente et al., 2005a). Waves

generally approach the coast from the west, however, southwest waves typically have greater importance during storms (Del Río et al., 2012). Despite the highest waves in the Atlantic reaching significant heights of up to 4m, over 70% of annual waves are less than 1m high, classifying the southern region of Spain as a low-energy coast (Benavente et al., 2000). In the southern Spanish coast, reduced westerly waves, lower sediment supply, and the presence of headlands cause an important decrease in the efficiency of the longshore component of sediment transport by waves. Coastal settings, including nearshore coastal orientation and bathymetry, which is the study of water bodies and their depths, also shape shoreline evolution.

Along the southern coast, rocky shore platforms modify diffraction/refraction wave patterns that can generate erosional hotspots (Kraus and Galgano, 2001, Anfuso et al., 2008)(Shoreline change patterns in sandy coasts. A case study in SW Spain).

Climate change projections indicate extensive and potentially catastrophic risks for ecosystems, human populations, built infrastructures, and economies of coastal areas, in particular for Southern Spain, port infrastructures, shipping operations, industrial zones, and surrounding natural habitats such as beaches and marshlands (Pérez, 2020).

2.1.2 Beach Erosion in Algeciras

Beach erosion is a prevalent issue along shorelines and refers to the loss of sediment from coastal areas, exceeding sediment gain. This imbalance, known as coastal sediment balance, is affected by factors such as sediment input, output, and transit changes. Sediments can be shifted alongshore by coastal currents or offshore by transversal currents. The decrease in sediment volume is a complex process that can cause shoreline recession (Prieto-Campos, et al., 2018).

The study “Methodology for Improving the Analysis, Interpretation, and Geo-Visualisation of Erosion Rates in Coastal Beaches—Andalusia, Southern Spain” was developed as a website using ArcGIS 10.3 software as well as the Digital Shoreline Analysis System tool (DSAS) to calculate erosion rates to see the changes in shoreline evolution of the entire Andalusian coast from 1956 to 2011. Below is a comprehensive picture of the most recent shoreline evolution in the Andalusian coast.

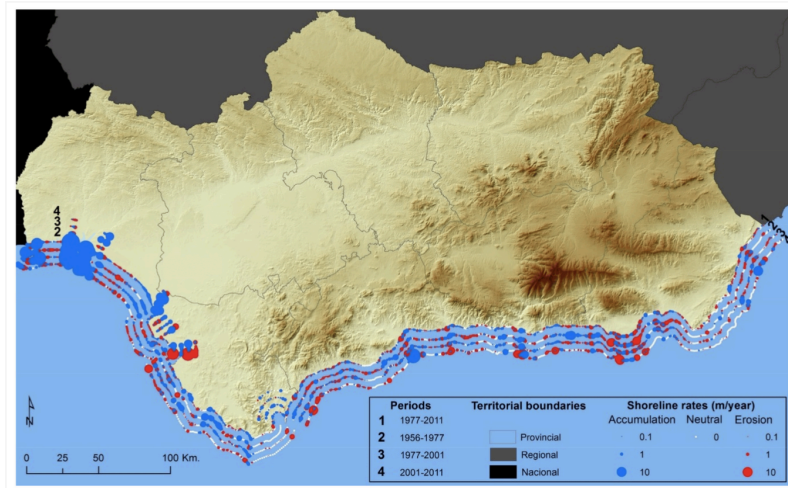


Figure 1: Erosion Rates Map of the Andalusian coast.

Reproduced with permission (Prieto-Campos, et al., 2018).

Algeciras, not as touristy as Cadiz or other famous seaside areas in Spain, have already suffered the loss of two beaches: El Chorruelo and Los Ladrillos. The community is currently risking the loss of another beach called El Rincocillo due to the expansion works carried out at the port of Algeciras (Pérez, 2020).

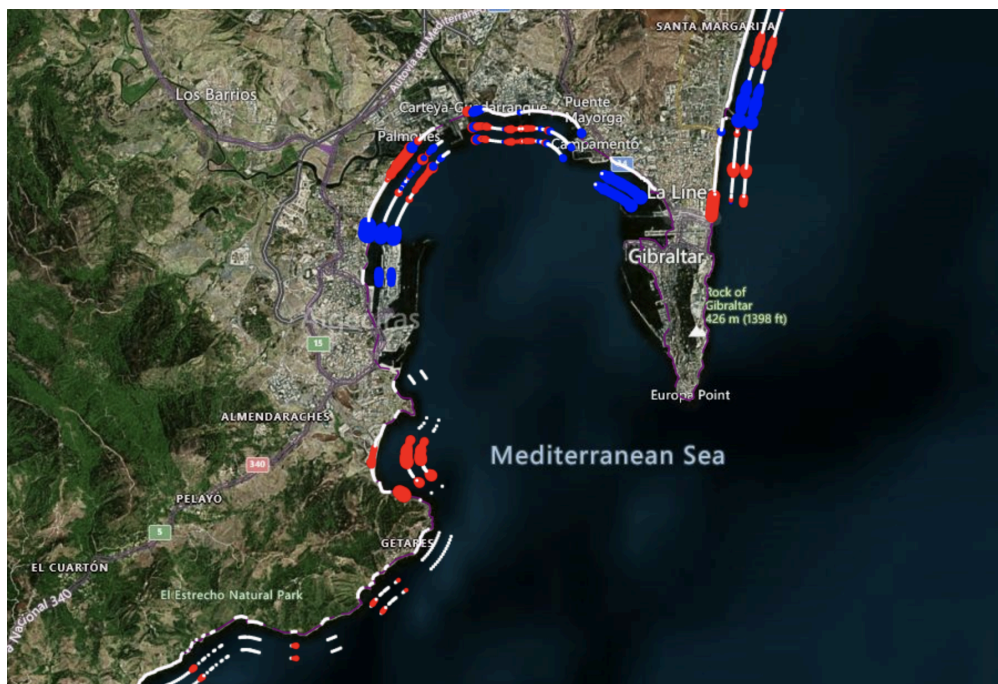


Figure 2: Erosion Rates Map of the Algeciras.

Reproduced with permission (Prieto-Campos, et al., 2018).

This issue arises because, as stated in the technical report directed by Pepe Cano, if the Juan Carlos I terminal had not been constructed, “the water would be able to continue entering into the Bay in the same way it previously had, pushing sand back and forth without any obstructions through a natural movement of currents and tides that stabilized the beach” (Pérez, 2020).

The report also states that the port and the extended constructions have lead to a phenomenon known as “later diffraction”, i.e. the tide hits an obstacle and a reflex effect is caused which generates a curve, which, “no matter how much sand is pumped into the beach, goes all the way down to the southern end, under the bridge”.

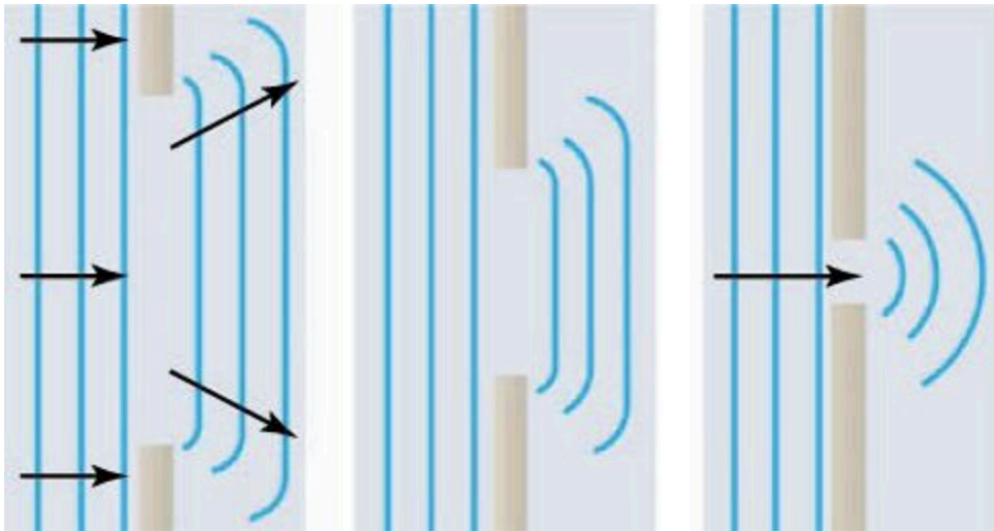


Figure 3: Wave Diffraction.

Reproduced with permission (Texas Gateway, 2024).

To face this challenge a group called ‘Salvemos Rinconcillo’ made up of concerned locals, ecologists, and people who love this beach in Algeciras are currently working to implement an underwater support structure to be installed in the bay to stabilize the coastline. This beach has continuously battled the changes in coastal dynamics for the last 20 years, and, specifically due to the construction and expansion of the container port, as it is built in the Bay, “no corrective environmental measures have since been taken”, reports the spokesman of the group, José Manuel Serrano. Serrano also stated: “This is a threat which the El Rinconcillo beach has been facing for a long time, but the damage is now so

evident that we have decided to get together and fight for our beach and for the Palmones marshland, which is also in danger of destruction... The recent storms have proven just how problematic the situation is, and we can't wait any longer" (Pérez, 2020).

The "Methodology for Improving the Analysis, Interpretation, and Geo-Visualisation of Erosion Rates in Coastal Beaches—Andalusia, Southern Spain" study results showed a rise in shoreline segments influenced by urban areas, increasing vulnerability to erosion. The Mediterranean coast exhibited higher vulnerability compared to the Atlantic coast, with 81% of stable segments. However, retrospective analysis revealed differences, with the Atlantic coast showing stability in some segments over time, while the Mediterranean coast experienced significant changes due to urbanization. This dynamic indicates a need for nuanced assessment, as relying solely on infrastructure proxies may lead to incorrect evaluations of beach stability, especially in the Mediterranean where increased rigidity limits natural variability (Prieto-Campos, et al., 2018).

As discussed, natural factors are not the only elements influencing shoreline evolution, but also human activities such as intense infrastructure and coastal engineering construction development, exert a significant influence on coastal erosion and accretion patterns. Coastal and beach erosion may induce a wide variety of negative ecological and socio-economical impacts such as increased loss of property and potential loss of life, damage to infrastructure, loss of tourism and recreation, contamination or disappearance of water storage wells, loss of coastal and benthic flora and fauna.

2.1.3 Dunes in Algeciras

Activities and infrastructures related to tourism and other human developments too (e.g., fishing and industrial activities) are significantly affected by the impacts of storms and hurricanes that, over the past century, have caused huge economic losses along with high mortality rates along the world's coastlines. Coastal erosion and flooding processes have reduced beach and dune ridges' width and produced the loss of associated touristic, aesthetic, and natural values (Rosa Molina, 2020).

The sand dunes that fringe the Spanish coastal landscape also serve an invaluable practical purpose. They act as a resilient barrier to the destructive forces of waves and wind. Dunes absorb the impact of high-energy storms preventing or delaying intrusion of waters into inland areas. Beaches and dunes are integral parts of a dynamic cycle in which sand is

constantly exchanged. During high-energy storms waves flatten the beach and erode sand, undermining and collapsing the seaward dune. On the other hand, the calmer waves carry sand from offshore bars and the surf zone to the beach, causing the beach to gradually accrete (Gregorio Gómez-Pina, 2022).

The location, development, and evolution of the coastal dune systems of the Cadiz coast are highly conditioned by the meteorological and climatic factors prevailing in the area. Climate changes, even the slightest ones, that have taken place throughout historical times, have affected the location of high and low atmospheric pressure systems in the North Atlantic, and, therefore have controlled and continue to control the direction and intensity of the dominant winds in the Gulf of Cádiz (F. Gabriel Garcia, 2011).

According to a document published by ‘Salvemos Rinconcillo’, the beach is continuously battling the changes in coastal dynamics dating back 20 years. Rinconcillo is home of a bunker constructed by the Franco-regime in the early forties against a possible Allied invasion. The bunker, which once was about 80 meters from the shore, is now in the sea most of the time, cutting off the beach (Rosario Pérez, 2020).



Figure 4: El Rinconcillo Bunker Algeciras in 2008 (Rosario Pérez, 2020).



Figure 5: Waves crash against El Rinconcillo Bunker Algeciras in 2019 (Rosario Pérez, 2020).

Past the bunker, once situated at a safe distance from the shore but now anchored at its base, stretches a line of dunes extending towards the river mouth. Experts caution that these dunes are diminishing, underscoring their ecological significance and crucial role in safeguarding the Palmones marshes. Carmen Fajardo, a professor and member of the Institute of Campo Gibraltarian Studies and the Cádiz Society of Natural History, expresses deep concern: "The current state of the dunes is alarming." She assures that "the visible erosion of sand due to recent storm activity" is leading to the degradation of this dune system, resulting in reduced sand volume, surface area, and height in certain regions. (Las dunas de El Rinconcillo, en peligro, 2020)

2.1.4 Strategies to Balance Ecological Preservation and Coastal Development

Spain, a global leader in coastal protection and climate adaptation, faces a crucial challenge: balancing essential urban development with the need to safeguard its unique marine ecosystems. Striking this balance is essential for protecting biodiversity, supporting coastal communities, and adapting to a changing climate (Muñoz Pérez, J. J., et al., 2021). Our sponsors authored strategies that incorporate these balances in a research analysis paper. The strategies can be divided into three categories: social and institutional prevention

measures, physical and structural prevention measures, and recovery and review of coastal protection.

Social and institutional prevention measures aim to enhance coastal protection through improved information, public awareness, and governance, through the use of hazard mapping, systemic coastal monitoring, and communication campaigns (Muñoz Pérez, J. J., et al., 2021).

Physical and structural prevention measures aim to mitigate erosion impacts by restoring or mimicking natural processes. These include coastal sediment management, which rebalances sediment by transferring it from accumulation to deficit areas, while beach regeneration involves reprofiling and constructing berms and artificial dunes. Dune system maintenance stabilizes dunes through vegetation replanting or invasive species removal, and marsh maintenance restores ecological processes by eliminating invasive vegetation and anthropic elements. Coastal stabilization relies on structures like breakwaters and dikes while containment structures such as concrete cladding and asphalt are utilized for coastal defense and upkeep of rigid structures to protect against coastline erosion, such as in the case of the beach of El Rinconcillo (Muñoz Pérez, J. J., et al., 2021).

Recovery and review efforts concentrate on tackling erosive events and enhancing coastal protection, functioning locally to restore affected areas while analyzing causes and consequences to prevent future damage. Evaluation, analysis, and diagnostics measures deepen comprehension of coastal system functionality and protection methods through close examination. These methods encompass analyzing past events, collecting data on affected populations, studying marine dynamics, and assessing vulnerability factors. Improved reconstruction efforts to diminish coastal erosion risk by leveraging erosive events as catalysts for essential transformation rather than solely focusing on restoration (Muñoz Pérez, J. J., et al., 2021).

2.1.5 Shortcomings of Current Coastal Management Strategies

Though the team, through interviews, intends to delve more into the overall strategies regarding current coastal management strategies, some of its shortcomings are well documented. Among the different strategies, the Salvemos Rinconcillo group opposes the "soft solution" of beach nourishment, which involves importing sand from elsewhere to

replenish lost sand. They argue that this method not only fails to address the issue effectively but also exacerbates the ongoing alteration of the area's characteristics because replenishing beaches with sand can be expensive and requires ongoing renourishment cycles due to natural processes. (Pérez, 2020). Specifically, they note that the imported sand tends to accumulate at the southern end of the beach, resembling Los Lances, while the northern end of El Rinconcillo deteriorates further.

The abrupt uprising of the El Rinconcillo group highlights the need for a more participatory and collaborative approach towards coastal management. When communities are not involved or informed of the decision-making process, they may feel disenfranchised and lack ownership over the implemented solutions (Kearney, et. al, 2007). Moreover, inadequate involvement could potentially lead to solutions that do not address the needs and concerns of local stakeholders (Muñoz Pérez, J. J., et al., 2021). This can lead to resistance and hinder the long-term success of implemented solutions.

2.2. Nature-Based Solutions(NBS): A Sustainable Approach

2.2.1 Defining Nature-Based Solutions

Nature-based solutions (NBS) offer a promising approach to addressing environmental and societal challenges in Algeciras, Spain. As defined by the International Union for Conservation of Nature (IUCN, 2023), NBS involves protecting, restoring, and sustainably managing natural and modified ecosystems to provide simultaneous benefits for both people and the environment. In the context of Algeciras, NBS can be employed to address pressing issues like coastal erosion, biodiversity loss, and promoting sustainable development.

A crucial feature of NBS lies in its ability to leverage the natural functions of ecosystems to solve environmental problems (World Bank, 2022). Dunes, for example, naturally absorb wave energy, offering a defense against coastal erosion, while wetlands act as natural filters for water purification (Rosa Molina, 2020). This highlights the

multi-beneficial nature of NBS, where a single intervention can address multiple challenges simultaneously. For instance, restoring dunes not only protects the coastline but also provides crucial habitat for diverse plant and animal life (World Bank, 2022).

Furthermore, NBS is designed with sustainability and minimal environmental impact in mind. They often involve long-term planning, stakeholder commitment, and ongoing monitoring to ensure their effectiveness (World Bank, 2022). Additionally, public awareness and education are vital for the long-term success of NBS, as community buy-in is essential for their implementation and maintenance (UN Environment Programme [UNEP], n.d.). Finally, for optimal effectiveness, NBS should be integrated with existing coastal management policies and plans to ensure complementarity and alignment (The Nature-Based Solutions Initiative, n.d.).

2.2.2 Role of Coastal Flora in Beach Protection

With the increasing damage of beach erosion, more innovative means to curb its effects have been sought. A strategy currently gaining traction has been the use of flora as protection (Davis, J.H, 1975, van Loon-Steensma et. al, 2017). The flora through their intrinsic properties like their long roots, provide ecosystem balance and fortification to the soil. This section delves into specific ways coastal flora could contribute to erosion mitigation, highlighting their invaluable role in beach protection.

One method of contribution is dune stabilization. According to the National Geographic Channel, a dune is a mound of sand formed by the wind, usually along the beach or in a desert. They grow as grains of sand accumulate, with every dune having a windward side and a slip face. In practice, they act as a first defense against frequent wave tides and forces. However, relentless assault from waves breaks them down leading to increased erosion (Feagin, R.A, 2019).

In general, once the main causes for dune degradation are either eradicated or mitigated, "soft" techniques, such as sand trapping, transplanted vegetation, fencing, dune walkovers, and environmental dune information dissemination, seem to be very efficient against dune degradation (Gregorio Gómez-Pina, 2022).

Research has proven that coastal flora can increase the resilience of the dune systems. In a Pilot study test result, Jacob B. Sigren (2014) ran streams of water for 45

seconds with constant velocity configurations through soils with and without *Sporobolus virginicus*, a perennial dune grass. Results showed that the shear strength of each root to withstand the velocity was significantly higher with soil with mature plants than bare soils.

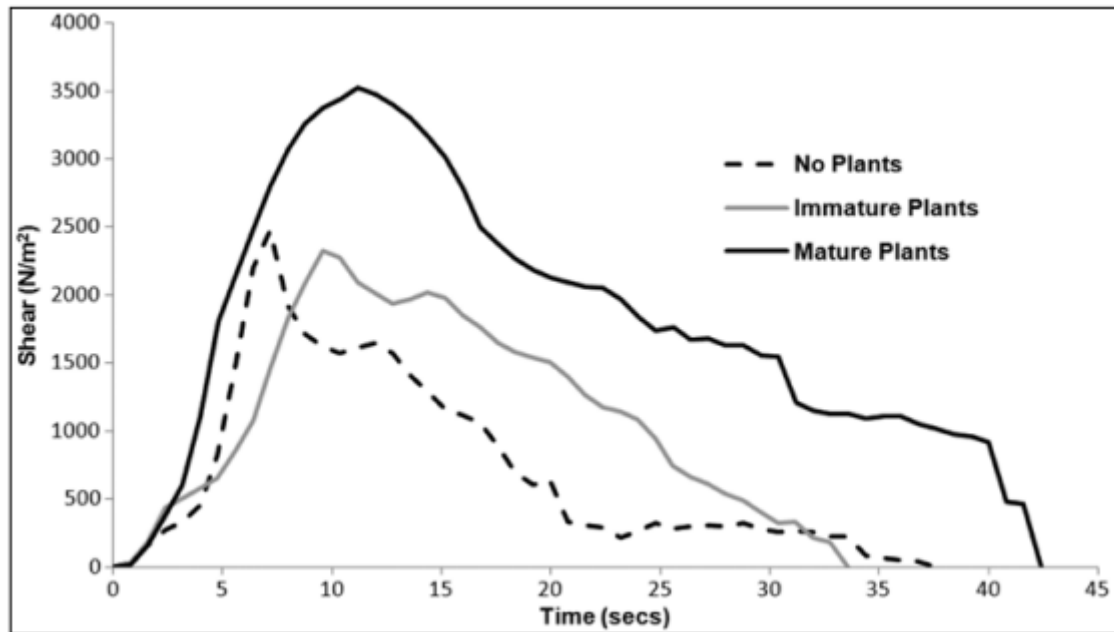


Figure 6: Soils containing varying levels of *Sporobolus virginicus* growing in dunes are subjected to constant velocity. (Sigren Jacob, 2014)

The reason for this is that the anchoring prowess of coastal plants lies in their extensive and deep root systems. The plants delve deep into the sand, forming a complex network that grips individual sand particles and binds them together. This intricate web acts like a natural geotextile, preventing wind erosion and resisting the shearing forces exerted by waves during storms (Bertoni et al., 2019).

The effectiveness of dune vegetation in combating coastal erosion is exemplified by the success story of Papamoa, New Zealand. Facing severe erosion in the early 1990s, the community launched a restoration program in 1994, focusing on revitalizing the seaward dune face with thousands of native plant species (Jenks & Brake, 2001). This initiative proved remarkably successful, achieving complete dune restoration and creating a wider, gentler slope with a robust, vegetated front that advanced by 10-25 meters, offering a natural buffer against future erosion.

Additionally, Figlus conducted an experimental study to investigate the physical support phenomena of flora roots. He analyzed to what extent vegetation increases the

capability of coastal dunes to resist wave-induced erosion. The experiments use the live dune grass species *Sporobolus virginicus* in wave testing. An important parameter in the testing was the plant's maturity and root system. The team ran four tests "consisting of 12 identical five-minute wave bursts for a total of 60 minutes of wave action per test". The results demonstrated that vegetated dunes are affected by erosion from both above- and below-ground variables. The above-ground parameters include plant height, stem width, canopy density, biomass, and rigidity, while the below-ground parameters include the plants' root size, depth distribution, density, and strength. These below-ground variables were found to be more influential in the dune morphology. Plant maturity and more developed root systems were found to have higher resistance to shear testing. Densely planted vegetation reduced reflected wave energy the most, in addition to slowing the dune scarp advancement rate, and reducing the total eroded dune volume by 8% in the irregular case and 30% in the regular wave case (Figlus, 2014). This study shows coastal morphology is greatly impacted by vegetation growth. Offers dune and landward infrastructure protection from storms and flooding, events that only increase in frequency due to the severity of climate change.

In addition to the more visible effects, coastal flora contributes significantly to the enrichment of soil organic matter through the decomposition of their leaves, stems, and roots. This organic material provides essential nutrients for various soil organisms, fostering a diverse and thriving microbial community. These organisms, in turn, further break down the organic matter, releasing vital nutrients like nitrogen, phosphorus, and potassium in forms readily available for plant uptake (Lodha, 2012). The presence of organic matter plays a crucial role in soil aggregation, the process by which individual soil particles clump together to form larger structures. This aggregation improves both soil aeration and water-holding capacity. A well-aerated and water-retentive soil creates a more favorable environment for plant growth, while also enhancing the soil's resistance to erosion from wind and wave currents (Merckx et al., 2001).

2.2.3 Flora Biodiversity in Algeciras

Andalucia is a unique location as two water bodies meet at the Strait of Gibraltar, producing substantial biodiversity. A notable place to witness this biodiversity is Parque

Natural del Estrecho, the southernmost protected area in Europe, with 18,931 hectares (73 square miles). Habitats in this region range from dunes and offshore sandbanks south of Algeciras to woodlands west of Bolonia (William, 2023 January). The Playa de los Lances, a beach within the Estrecho National Park, has a range of flora comprised of sea holly, sea daffodils, and southern birdsfoot-trefoil. Sea holly and Marram grass are known to have properties important to reinforce sediment and dune formations. As this beach is exposed to and known for strong winds, access can be challenging due to the severity of windswept sediment along the coastline (Williams, 2023 April).

Relevant flora in Algeciras has been identified to contribute to nature-based solutions to counter the effects of beach erosion. A nature-based solution is a sustainable intervention, powered by nature, placed into the environment to develop adaptation and resilience, commonly towards climate change, dune restoration, and flood risks. Extensive research by the International Union for Conservation of Nature has been done on the seagrass species *Posidonia oceanica*. Commonly known as Neptune Grass, the beds reduce the speed and strength of waves, helping to prevent destructive flooding events. Neptune Grass also has remarkable carbon-sequestering properties, estimated to conceal 10% to 40% of the Mediterranean's greenhouse gasses since the Industrial Revolution (Interreg Mediterranean). Two beach grass species also found in Algeciras have extensive root systems that flourish in sandy areas, *Ammophila arenaria* and *Eryngium maritimum*. These two beach grasses are commonly known as Marram Grass and Sea Holly (GISD, 2010).

2.2.4 Global Success Case Study: North Brittany, France

The dunes of North Brittany, France accounted for 13% of its coast (Roze, 2004). Once a source of pride and protection, the dunes faced a grim reality by the mid-20th century. The reasons for this were largely because increased development was not equally matched by nature preservation methods. Overgrazing by sheep prevented the growth of stabilizing vegetation, leaving the dunes vulnerable to windblown sand erosion. Unsustainable sand extraction further destabilized the dunes by disrupting the region's natural sand accumulation processes. Increased tourism added to the problem because it brought uncontrolled foot traffic and recreational activities that compounded the erosion.

The consequences were starkly evident. Dunes lost up to 80% of their height and volume, jeopardizing their ability to shield inland areas from storm surges and flooding.

This erosion also negatively impacted the ecological health of the region, as degraded dunes offered limited habitat for coastal wildlife.

In response to this critical situation, restoration projects were initiated in the 1970s. These projects embraced a nature-based solution, focusing on the planting of native *Ammophila arenaria*, also known as sand couch grass (Roze, 2004). This choice proved successful due to the grass's unique characteristics. Its long, sand-trapping stems and rhizomes effectively capture windblown sand particles, promoting dune formation. Sand couch grass thrives in the harsh coastal environment, withstanding strong winds and salt spray. Additionally, its readily available nature and fast-growing characteristics made it a cost-effective option for large-scale restoration efforts.

The restoration efforts employed various techniques, including direct seeding of sand couch grass in prepared areas, often following specific patterns to maximize sand trapping. In some cases, pre-grown seedlings or "plugs" were planted to accelerate the establishment of the grass cover. To protect newly planted areas and prevent further disturbance, temporary fencing was often installed, restricting public access and allowing the grass to be established. In critically eroded areas, sand nourishment was sometimes employed, involving the addition of sand to facilitate dune rebuilding.

These efforts yielded significant and measurable improvements. Studies suggest that sand couch grass planting resulted in reductions of up to 70-80% in erosion rates compared to unplanted areas. The restored dunes regained their natural form, with documented increases in height and volume in various locations. This translated to enhanced coastal protection, as the restored dunes now provide improved protection against storm surges and flooding, safeguarding inland areas and infrastructure. A monitoring study conducted by Francoise Roze (2004) confirmed the re-accumulation of sand by dunes post-flora planting efforts. Results proved there was up to a 2m increase in the rear face of the dune.

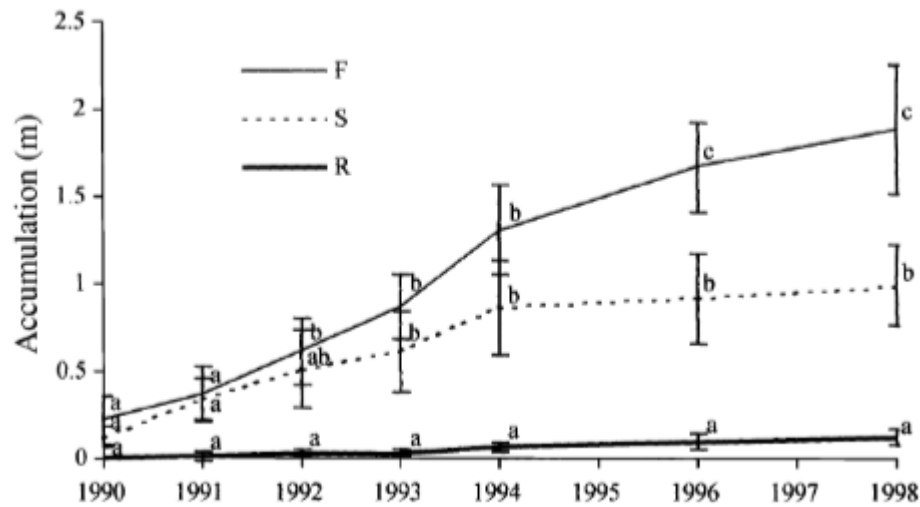


Figure 7. Evolution of sand accumulation in the front dune (F), the dune summit (S), and the rear dune (R). (Roze, 2004)

The North Brittany dune restoration project serves as a powerful example of the effectiveness of flora as a nature-based solution in addressing coastal erosion. It highlights the effectiveness of stabilizing dunes and restoring ecological functions while emphasizing the importance of ongoing monitoring and adaptation in the face of evolving environmental pressures.

2.2.5 Existing Projects in Protecting Native Biodiversity

Current research on climate change impacts highlights the crucial role of natural ecosystems in human well-being and global processes. The necessity of protecting these ecosystems is highlighted by the Convention of Biological Diversity which emphasized the need to mitigate climate change and promote adaptation (Roberts et al., 2020). Targeting the end of 2030, the Convention of Biological Diversity proposed increased protection in areas both on land and in the sea (Roberts et al., 2020). These protected areas can serve as important tools for achieving an understanding of climate change mitigation and

biodiversity conservation. However, the success of these areas is dependent on location, public engagement, and governance (Roberts et al., 2020).

Various strategies and organizations work to combat the loss of biodiversity in Spain. One strategy is through Marine Protected Areas (MPAs), which carry out conservation programs generated by the Convention on Biological Diversity and the European Union (Alcorlo et al., 2023). MPA works to regulate allowed fishing activities in particular areas, limiting the exploitation of marine life by fisheries to focus on the conservation of such species (Alcorlo et al., 2023). The Spanish Society of Plant Conservation Biology (SEBiCoP) was created in 2003 by a group of botanists and conservationists in Spain who work to identify and evaluate wild plants to determine appropriate conservation management (Moreno Saiz et al., 2018). The SEBiCoP implements methods such as restoration through the propagation of new seedlings, translocation of flora, or the reinforcement of areas known to house endangered species (Moreno Saiz et al., 2018). As well, SEBiCoP utilized methods such as ex-situ conservation of plant seeds to combat extinction (Moreno Saiz et al., 2018).

Sponsor

The sponsor for this project is Antonio Contreras De Villar whose current research consists of developing and evaluating “Strategies for coastal protection in the provinces of Cádiz, Málaga, and Almería considering the effects of climate change”.

This research aims to gather sufficient data to understand the medium and long-term risks and the resilience of the Andalusian coastline, as the ongoing rise in sea levels, it is anticipated that there will be a rise in storm frequency and wave intensity, leading to heightened coastal flooding and erosion. The subsequent impacts on port infrastructure, shipping operations, industrial zones in the vicinity, and surrounding natural habitats such as beaches and marshlands are significant but currently uncertain. As they are currently working on the proper implementation and development of the strategies discussed in

section 2.1.3, the team hopes to add to their current research the incorporation of nature-based solutions into existing practices.

3. Methodology Chapter

The goal of our project is to explore the feasibility of flora as a nature-based solution to combat beach erosion in Algeciras. The team intends to understand the importance of biodiversity in Algeciras, the current practices to combat beach erosion, the limitations and challenges around these practices, how flora can be used as nature-based solutions to combat beach erosion, and how these nature-based solutions can be integrated into existing practices.

The team has identified the following objectives to guide our research.

- Document and understand the flora presence in the study area
- Identify current practices and policies to address beach erosion
- Identify strategies to incorporate nature-based solutions into existing practices

3.1.1 Document and Understand the Flora Presence in the Study Area

To achieve this objective the team will conduct transect walks with local experts, environmental agencies and botanists such as the Biodiversity Foundation (Fundación Biodiversidad), the Agencia de Medio Ambiente y Agua de Andalucía (AMAA), The Spanish Society of Plant Conservation Biology (SEBiCoP) and LifeWatch ERIC, to identify the flora species and presence across our study area in Algeciras. The team plans to conduct this transect walks both on beaches in the Bay of Algeciras past Juan Carlos I container port such as El Rinconcillo beach which is more affected by the port development, and before the port, such as Playa de Getares which should be in similar settings as El Rinconcillo beach but shouldn't be as much influenced by the port development.



Figure 8: Map of the Bay of Algeciras, the team's potential study area.

(Wikimedia Foundation, 2013)

The team intends to gather data on the composition and density of the flora population on the two beaches. The team will be taking pictures to identify the different species and doing transect sampling to identify the density of flora present, which consists of establishing transect lines across the beach (randomly or systematically) and, along each transect line recording the presence and abundance of flora at regular intervals.

3.1.2 Identify Existing Practices and Policies to Address Beach Erosion

To achieve this objective the team will conduct interviews with coastal management and environmental agencies to gain a better understanding of the current policies that are in place, the limitations of these policies and how nature-based solutions could be integrated to the current practices. Within the first weeks of the project, the team will set up 3-6 interviews with coastal management agencies such as Demarcación de Costas de Andalucía Atlántico, the Salvemos Rinconcillo group, and Port Authority of the Bay of Algeciras, along with environmental agencies such as the Biodiversity Foundation (Fundación Biodiversidad), the Agencia de Medio Ambiente y Agua de Andalucía (AMAA), The Spanish Society of Plant Conservation Biology (SEBiCoP) and LifeWatch ERIC. Through interviews with coastal management and environmental agencies, the team aims to gain a better understanding of their current consideration and general attitude toward the potential integration of nature-based solutions in the current practices and what are the major limitations and challenges around implementing nature-based practices. The interview protocol can be found at the Appendices.

3.1.3 Identify Strategies to Incorporate Nature-Based Solutions into Existing Practices

Finally, after obtaining and analyzing the data from the interviews, and the results from the transect walks, the team plans to conduct a focus group made of our sponsor, environmental agencies, university faculty, and interested stakeholders to evaluate the current policies that are in place. The team hopes to incentivize a broader discussion to

understand better if these current policies properly balance beach and biodiversity preservation and if these practices could be improved with the integration of nature-based solutions and the feasibility of these solutions to combat beach erosion.

3.1.4 Project Schedule

The team intends to conduct interviews with coastal management agencies throughout the first two weeks, to identify and understand their perspective, and to then conduct interviews with environmental agencies throughout the second and third week of the term. After having conducted the interviews, within the second and third week, the team intends to conduct transect walks with experts to identify the composition and density of the flora in the study area. Finally, the team will conduct a focus group with interested stakeholders and university faculty members to enhance a broader discussion on the topic with the hope of identifying ways to integrate nature-based solutions into the current policies.

Table 1: Project Schedule

Time	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7
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Location	Cadiz	Algeciras	Algeciras	Algeciras	Algeciras	Cadiz	Cadiz
Interviews with coastal management agencies	X	X					
Interviews with environmental agencies		X	X				
Transect walks		X	X				
Focus Group				X	X		
Final Presentation						X	X

Appendices

Interview Format

Date:

Time:

Location:

Attendee(s):

Interview with _____,

Introduction and Informed Consent:

Hello,

We are a group of university students from Worcester Polytechnic Institute (WPI) located in Massachusetts, USA. We are currently working on a project in collaboration with the University of Cádiz, aimed at understanding the impacts of coastal erosion on biodiversity in Algeciras and how nature-based solutions can be integrated into the current policies to combat coastal erosion. We are conducting interviews with coastal management and environmental agencies to learn more about the current practices that are in place in order to understand the main challenges among these practices and how nature-based solutions could be integrated. We intend that our research will contribute insights to implement further policies to combat beach erosion and mitigate loss of biodiversity.

Your participation in this interview is entirely voluntary and you are free to withdraw from the interview and discontinue our discussion at any time. Do we have your permission to record this interview and your answers to utilize in our report? If you prefer, we can ensure the confidentiality of your answers by excluding your name, and any personal information that may be associated with any of your responses. If you would like, we are willing to provide you with the results at the end of the project. Feel free to contact our team at Cadiz_24_Coastal_Structures@wpi.edu.

Signature of Participant

Date

Before we begin, do you have any questions or concerns you would like to share with us?

Questions:

CMA: Coastal Management Agencies;

EA: Environmental Agencies

Questions	CMA	EA
What are the current practices in place to combat coastal and beach erosion? Can you describe them?	X	X
Do you think that these practices are effective? Why?	X	X
How does the port authority currently address environmental concerns raised by Salvemos Rinconcillo arising from its operations near the coastal ecosystem?	X	X
What is currently being done to mitigate beach erosion and loss in biodiversity? Do these practices take into account the issue of loss in biodiversity?	X	X
Are you aware of any examples of successful nature-based solutions implementation for beach erosion control in other coastal regions? If so, can you name and explain them?		X
How do you envision the potential role of flora as a nature-based solution, particularly the use of coastal flora, in mitigating some of the environmental concerns associated with port operations and beach erosion?		X
What challenges do you foresee in implementing flora as nature-based solutions to combat beach erosion?		X
What do you think are the challenges around the implementation of these coastal management practices?	X	X
Is there anything else that you would like to mention that might be important for our project?	X	X

Thank you for taking the time to talk to us today! Are there any responses you wish to change or alter? If there is anything else you may want to add, feel free to contact our team at Cadiz_24_Coastal_Structures@wpi.edu.

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